

SHETH L.U.J. AND SIR M.V. COLLEGE

Aim:

Mote-to-PC Serial Communication Simulation

Implement serial communication between a sensor mote and PC to monitor real-time data transmission.

Code:

Mote.ned:

```
package pc;
```

```
simple Mote
```

```
{
    parameters:
        double sendInterval @unit(s);
    gates:
        output out;
}
```

PC.ned:

```
package pc;
```

```
simple PC
```

```
{
    gates:
        input in;
}
```

SerialNetwork.ned:

```
package pc;
```

```
network SerialNetwork
```

```
{
    submodules:
        mote: Mote
        {
            parameters:
                sendInterval = 1s;
        }
        pc: PC;
    connections:
        mote.out --> pc.in;
}
```

Mote.cc:

```
#include <omnetpp.h>
```

```
class Mote : public cSimpleModule
```

```
{
    private:
        cMessage *timer;
```

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protected:

```
virtual void initialize()
{
    timer = new cMessage("timer");
    scheduleAt(simTime()+1, timer);
}

virtual void handleMessage(cMessage *msg)
{
    if(msg == timer)
    {
        cMessage *data = new cMessage("Temperature=25C");

        EV << "Mote Sending Data\n";

        send(data, "out");

        scheduleAt(simTime()+1, timer);
    }
}

virtual ~Mote()
{
    cancelAndDelete(timer);
}
};

Define_Module(Mote);
```

PC.cc:

```
#include <omnetpp.h>

class PC : public cSimpleModule
{
protected:

    virtual void handleMessage(cMessage *msg)
    {
        EV << "PC Received: "
            << msg->getName()
            << "\n";

        delete msg;
    }
};
```

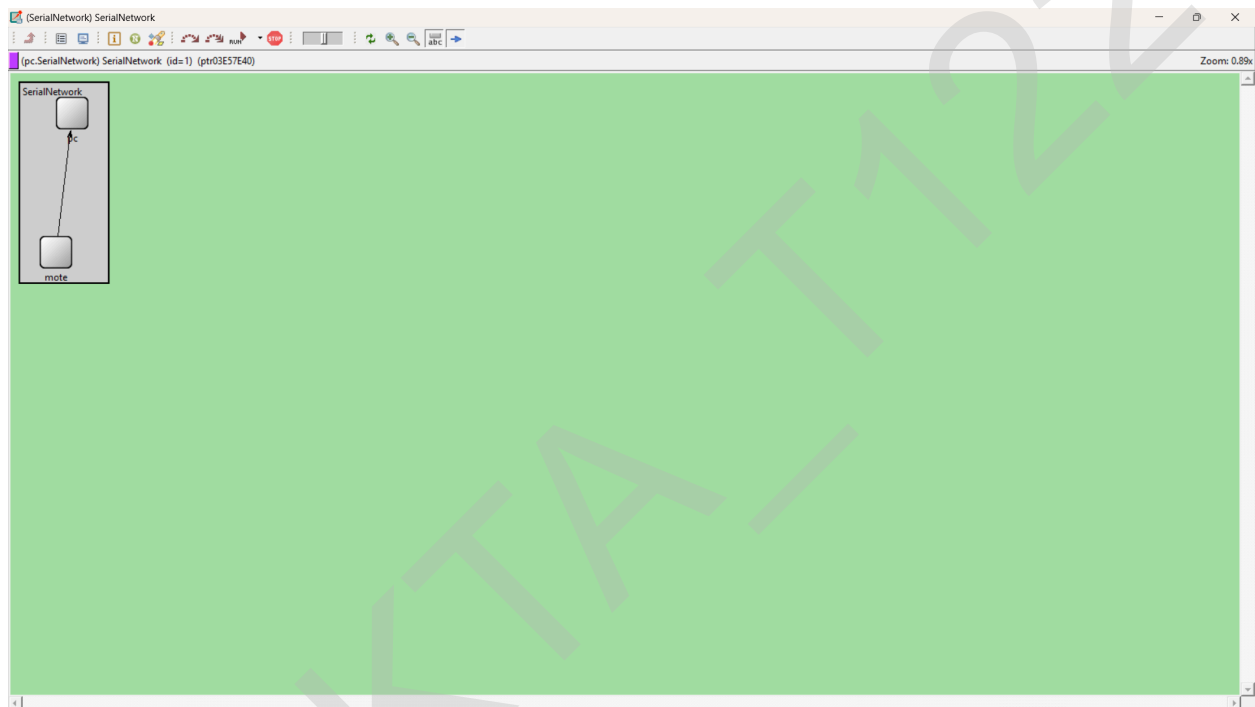
```
}  
};  
Define_Module(PC);
```

Omnetpp.ini:

```
[General]  
network = pc.SerialNetwork  
sim-time-limit = 20s
```

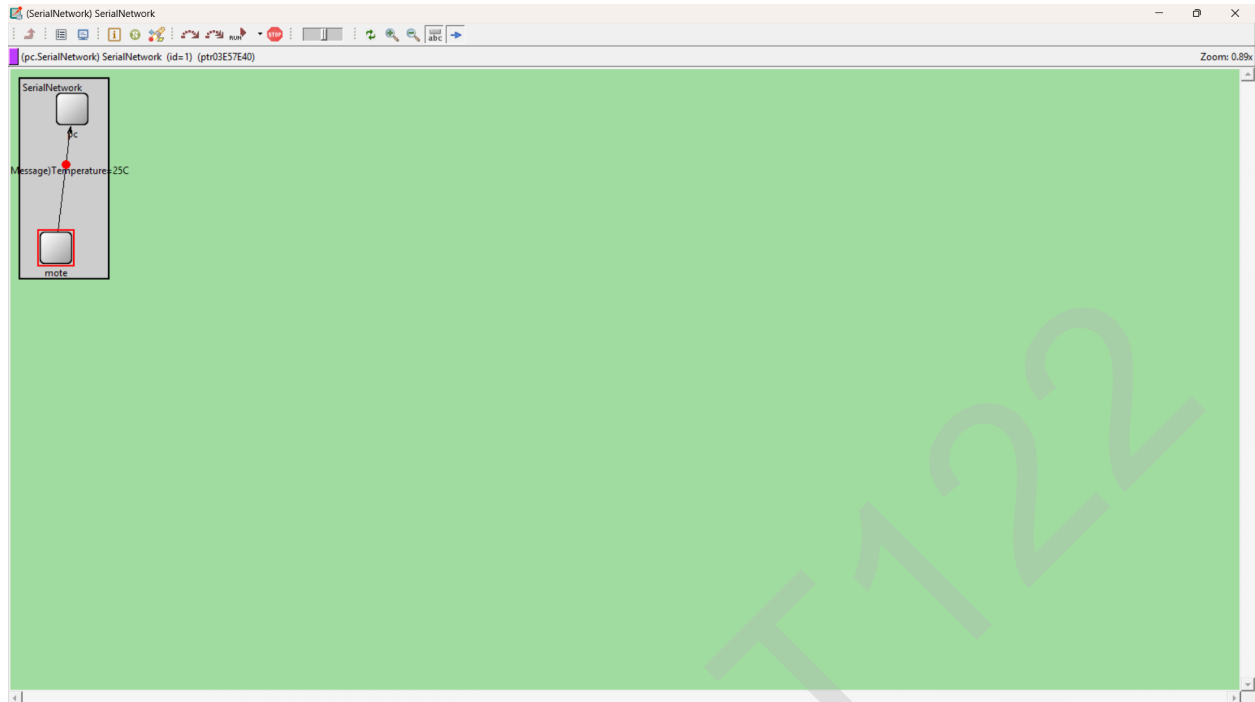
Output:

Serial Network Topology

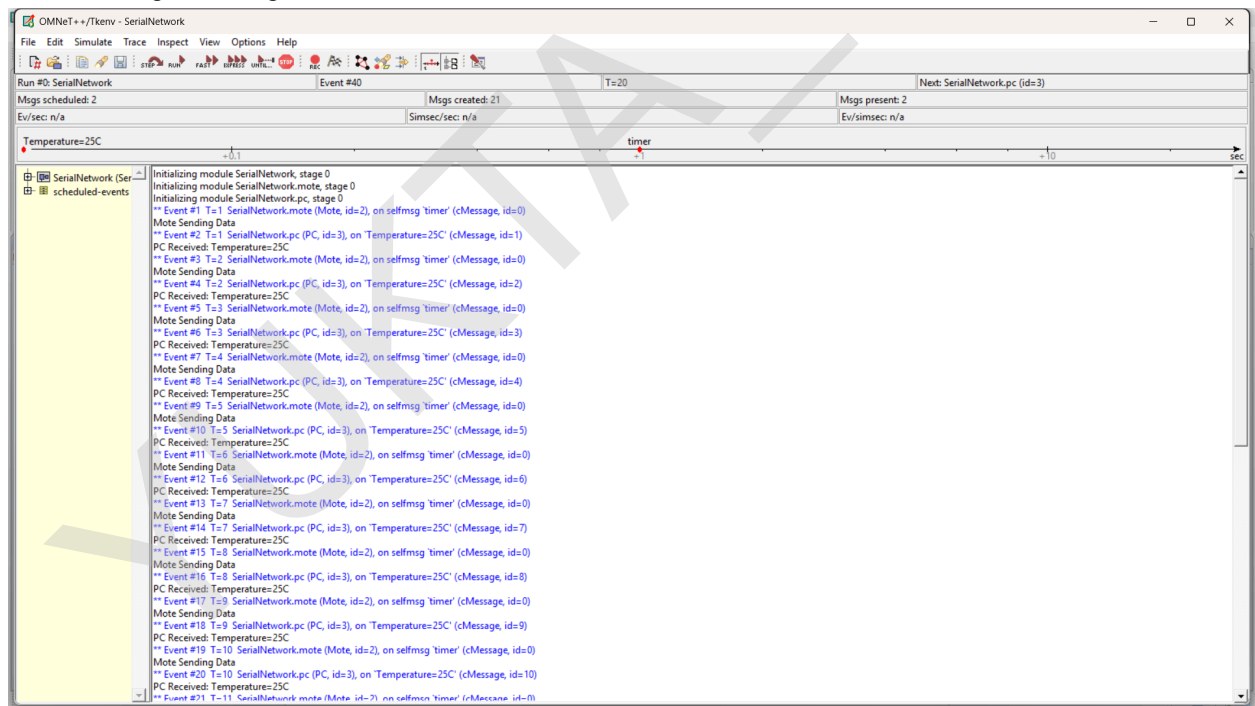


Simulation Execution:

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Mote Sending Data Log
PC Receiving Data Log



Result

The sensor mote successfully transmitted data to the PC through a serial communication link. The PC received and displayed the transmitted data in real time during the simulation.